

Where Sampling-Grid Dependence Enters Selective State-Space Models

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Abstract

A sequence model can change its output when only the time grid of the same physical input changes. We ask where this dependence enters a selective state-space model (SSM), separating interval splitting, which keeps the held input path, from new observations, which change the input reconstruction; operationally, an artifact is an output change under an unchanged held path. For one layer, standard conditions (a physical step, exact zero-order hold, coefficients that see only the held value) give split invariance; in a stack, later layers receive an internal signal that the refined grid samples differently, and for a scalar linear cascade we give an $O(\text{mesh})$ upper bound on the resulting error. In the tested single-layer toy ablation, the combined corrections reduced the readout artifact to rounding level, whereas no single correction did. On the official code of one selective SSM with a physical step (TIDES), trained with four seeds on one synthetic regression task, we separate three effects. Model coupling: eight-fold splitting of the held input changes the outputs by 4.3×10^{-3} –0.022 (relative ℓ_2) in every seed. Pooled weighting: under a same-path density change the token-mean readout moves by +0.46 to +0.50 label units on average, of which +0.48 to +0.49 is the weighted-average identity $\frac{7}{18}(A - B)$ of the unchanged outputs rather than a property of the SSM; time weighting of the same outputs leaves a mean absolute change of 0.013–0.023. Task risk: the native-versus-resampled loss contrast changes sign across seeds, while resampling is about ten times cheaper at eight-fold refinement. We propose no new architecture.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Continuous-time state-space layers are applied to sequences by discretizing a linear ODE over each interval between observations. When the step is the physical elapsed time, the recurrence inherits the semantics of the continuous system, which lets S4, LSSL and S5 change the sampling rate at test time or consume irregular timestamps (Gu et al., 2021a;b; Smith et al., 2022). Selective SSMs such as Mamba compute the step from the current token (Gu & Dao, 2023; Dao & Gu, 2024), so it acts as a content gate; TIDES restores a physical step and moves selectivity to the state matrix (Soydan et al., 2026).

We do not propose another architecture. We ask a measurement question: *when only the time grid of the same physical input path changes, where does the output change enter?* This requires separating two operations. *Splitting* a held interval into m sub-steps that repeat the held value leaves the held input path unchanged; we call an output change at the common times an *artifact*, in the operational sense of a violation of this pre-declared invariance. *Adding observations* of the same signal changes the observed values and hence the reconstructed input path, so part of the change is a response to a different input; we make no claim about task-relevant or Bayesian information.

Known and not claimed. The physical Δt in the discretization, exact zero-order-hold (ZOH) formulas and flow composition are standard (Gu et al., 2021b; Smith et al., 2022); that the released Mamba code uses $\bar{B} = \Delta B$ (“exponential-Euler”) is documented by Lahoti et al. (2026); that a continuous-time reading of selective SSMs needs $\Delta \propto dt$ is shown by Cirone et al. (2024); keeping Δ physical with selectivity on the decay rate is the design of Soydan et al. (2026); and that a sample mean weights samples rather than time is elementary.

Contributions. None is an architecture.

1. A protocol that separates splitting from new observations, with an exact path/artifact decomposition and explicit denominators (Section 3).
2. Two different causes, stated with their assumptions: resampling of internal signals between layers, which

changes a stack of exact-update layers under splitting although the held input is unchanged, with an $O(\text{mesh})$ upper bound for a scalar linear cascade (Section 4.2), and sample-count readouts, which weight tokens instead of time (Section 4.3); toy checks in Section 6.

3. A fixed-protocol seed repetition on the official TIDES code that separates a model-coupling effect, a pooled-weighting effect and a task-risk effect, with cost; an exact signed decomposition shows that the token-mean readout effect is almost entirely a weighted-average identity (Section 5).
4. A check that the TIDES paper equations and the released code pair inputs with different intervals (Section 4.4).

2. Related Work

Discretization in structured SSMs. S4 uses the bilinear transform and reports a test-time resolution change obtained by changing its internal step (Gu et al., 2021a); LSSL handles irregular data by discretizing with a different timescale (Gu et al., 2021b); S4D gives bilinear and ZOH rules (Gu et al., 2022). S5 uses ZOH, $\bar{\Lambda} = e^{\Lambda\Delta}$, $\bar{B} = \Lambda^{-1}(\bar{\Lambda} - I)\tilde{B}$, rescales Δ globally for zero-shot 8 kHz evaluation of a 16 kHz model, and supplies a per-step Δ_t for an irregular pendulum task, with ablations that drop the interval or append it as a feature (Smith et al., 2022). S4ND notes that only the relative rate matters (Nguyen et al., 2022); event-camera SSMs evaluate at different frequencies (Zubić et al., 2024). These are LTI layers.

Selective SSMs. Mamba makes Δ, B, C input-dependent and states the ZOH rule (Gu & Dao, 2023); Mamba-2 keeps its discretization (Dao & Gu, 2024). Mamba-3 derives right-endpoint ZOH for time-varying systems, names the released rule exponential-Euler, and proposes an exponential-trapezoidal rule that is second order only if its data-dependent weight is $\frac{1}{2} + O(\Delta_t)$ (Lahoti et al., 2026); its evaluation concerns language modeling and related tasks. Liquid-S4 makes the transition input-dependent (Hasani et al., 2022).

Physical time in selective SSMs. TIDES keeps $\tilde{\Delta} \equiv \Delta$, makes $\text{Re } \Lambda, B$ and C input-dependent, and uses $\bar{B}_k = \Lambda_k^{-1}(\bar{\Lambda}_k - I)B_k$ (Soydan et al., 2026). Its Fading Flash diagnostic rescales a global step for fixed-length sequences and shows that a Mamba surrogate with Δ as an input channel fails outside the training range; its random-drop study removes observations and reports accuracy. In the sections we read it neither refines a fixed held path nor decomposes output changes. It is the closest prior work; we evaluate its official code, not a re-implementation, and we treat its architecture as prior art.

Continuous-time models. Neural CDEs are driven by an interpolation of the observations (Kidger et al., 2020); Log-NCDEs build on NCDEs, which are robust to irregular sampling (Walker et al., 2024); Cirone et al. (2024) read selective SSMs as linear CDEs in a limit where δ plays the role of dt ; oscillatory SSMs study implicit and IMEX integration (Rusch & Rus, 2024). We did not find, in these sources, a refinement of the same held path for selective SSMs that separates splitting from new observations and treats readouts and inter-layer resampling as entry points; this is absence of evidence, not proof of novelty.

3. Problem Setup

Model and holds. The toy is a scalar-input selective model with N diagonal modes,

$$\dot{h}_n = g(u)(\lambda_n h_n + B_n(u)u), \quad y = \text{Re} \sum_n C_n(u)h_n, \quad (1)$$

with $\text{Re } \lambda_n < 0$, a gate $g(u) = \text{softplus}(wu + \beta)$ shared across modes, and affine $B(u), C(u)$. A grid $0 = t_0 < \dots < t_K = T$ has steps dt_k and mesh δ . We use the right-endpoint hold $\tilde{u}(t) = u(t_k)$ on $(t_{k-1}, t_k]$, which is the convention of the Mamba recurrence and of the TIDES code (Section 4.4). A discrete variant is $h_k = \bar{A}_k h_{k-1} + \bar{B}_k u_k$ with $\Delta_k = dt_k g(u_k)$ (“ $\Delta = dt g$ ”) or $\Delta_k = \tau g(u_k)$ with a fixed τ (“ $\Delta = \tau g$ ”), and exact ZOH, Euler- B ($\bar{B} = \Delta B$) or bilinear rules; τ is the base step.

Grid changes and decomposition. *Splitting* divides base intervals into m equal sub-steps that all receive the held value; *new observations* give each new sub-time the value of the continuous signal. Outputs are compared at the common base times. With D_m the outputs of a variant on grid m and Y_m the continuous-time solution of (1) on the held path of grid m , the change splits exactly as

$$T_m = \underbrace{(Y_m - Y_1)}_{P_m: \text{path}} + \underbrace{[(D_m - Y_m) - (D_1 - Y_1)]}_{R_m: \text{artifact}}, \quad (2)$$

so $\|T_m\|^2 = \|P_m\|^2 + \|R_m\|^2 + 2\langle P_m, R_m \rangle$. Under splitting $P_m = 0$. Because the cross term can be large and of either sign, we never report a ratio of norms as a share.

Readouts. We distinguish sample-count readouts ($\sum_k y_k$, $\frac{1}{K} \sum_k y_k$), time-weighted endpoint sums (right-endpoint $\frac{1}{T} \sum_k y_k dt_k$ and trapezoidal), and the time average $\frac{1}{T} \int_0^T y dt$ of the output over the held path, which the toy computes in closed form for exact-ZOH states (the “exact integral”).

4. Where Grid Dependence Enters: Analysis

4.1. Single layer

Proposition 4.1 (standard). *Fix a base interval $(t_{k-1}, t_k]$ and suppose (i) every sub-step receives the held value u_k ; (ii) every coefficient (g, B, C, λ) is a function of the held value only; (iii) the step is (sub-step length) $\cdot g(u_k)$; and (iv) $\bar{A}, \bar{B}$ are the exact ZOH formulas. Then, from the same state at t_{k-1} , the state and output at t_k are the same for every m ; by induction the outputs at all base times are split-invariant.*

Proof. Each sub-step is the exact flow Φ_s of $\dot{h} = g(u_k)(\lambda h + B(u_k)u_k)$ over its length s , and $\Phi_{s_1} \circ \Phi_{s_2} = \Phi_{s_1+s_2}$. $\square$

The proposition is textbook; its use is that each entry point violates one assumption. $\Delta = \tau g$ violates (iii): m sub-steps apply m times the decay. Euler- B violates (iv), with local error $|\Delta B u| |\varphi_1(\Delta \lambda) - 1| \leq |B u| |\Delta^2| |\lambda|/2$, $\varphi_1(z) = (e^z - 1)/z$, since $e^z - 1 - z = z^2 \int_0^1 (1-s) e^{sz} ds$ gives $|\varphi_1(z) - 1| \leq |z|/2$ for $\text{Re } z \leq 0$. (ii) fails whenever a coefficient sees anything besides the held value: a token-indexed convolution (Mamba’s `d_conv=4`), time or position features, state-dependent transitions, or training-mode normalization statistics over time. Bilinear is stable for $\text{Re } z < 0$ at every step, but for real $z < -2$ its transition is negative (sign-alternating transients) and it tends to -1 instead of $e^z \rightarrow 0$ (inaccurate stiff decay); its held-input fixed point is exact.

4.2. Several layers: internal resampling

In a stack, layer 2 is driven by the continuous output of layer 1, but a layerwise recurrence gives it only endpoint samples held over each step. Splitting the input grid changes which internal samples it sees, without adding any external observation. The minimal instance is

$$\dot{h}_1 = -a h_1 + b u(t), \quad \dot{h}_2 = -c h_2 + d h_1(t). \quad (3)$$

Proposition 4.2 (standard). *Let u be held on a grid with mesh δ , $|u| \leq U$, $a \geq 0$, $c > 0$ and $h_1(0) = h_2(0) = 0$. (a) The exact coupled solution of (3) at base times is split-invariant. (b) The layerwise computation that is exact in layer 1 and holds $h_1(t_k)$ on $(t_{k-1}, t_k]$ in layer 2 has absolute error $|e_k| \leq |b d| U \delta e^{c\delta}/c$ at every grid time, for every T and also for $a = c$.*

Proof. (a) follows as in Proposition 4.1, since the coupled system has constant coefficients on each held interval. (b) Layer 1 is exact, so with $\tau_k = dt_k$, $e_k = e^{-c\tau_k} e_{k-1} + d \int_0^{\tau_k} e^{-c(\tau_k-s)} [h_1(t_k) - h_1(t_{k-1} + s)] ds$. If $a > 0$ then $|h_1| \leq |b|U/a$, so $a|h_1| \leq |b|U$; if $a = 0$ then $a|h_1| = 0$; either way $|\dot{h}_1| \leq 2|b|U$. On each held interval

h_1 is C^1 , so the bracket is at most $2|b|U(\tau_k - s)$, and with $e^{-c(\tau_k-s)} \leq 1$ the local term is at most $|b d|U\tau_k^2$. Unrolling from $e_0 = 0$ gives $|e_k| \leq |b d|U\delta \sum_{j \leq k} \tau_j e^{-c(t_k-t_j)}$, and since $e^{-c(t_k-t_j)} \leq e^{c\delta} e^{-c(t_k-s)}$ for $s \in (t_{j-1}, t_j]$, the sum is at most $e^{c\delta} \int_0^{t_k} e^{-c(t_k-s)} ds \leq e^{c\delta}/c$. $\square$

Remark 4.3 (scope of Proposition 4.2). The argument is elementary and has not been checked by a third party. A nonzero $h_1(0)$ adds $a|h_1(0)|$ to $2|b|U$; a nonzero $h_2(0)$ does not enter. Uniformity in T needs $c > 0$: as $c \rightarrow 0$ the sum is bounded only by t_k and the bound grows linearly in time; $a < 0$ (an unstable first layer) is not covered. In the stored P1-COMP-01 outputs, the largest error at the base times is 9–20% of the bound for $m \in \{1, 2, 4, 8\}$, which is consistent with, not a proof of, the bound. The statement is for a linear time-invariant scalar cascade; it gives no guarantee for input-dependent coefficients, nonlinear maps between layers, or trained deep models, where the Lipschitz and contraction constants it uses are not controlled.

Conditions (ii)–(iv) of Proposition 4.1 concern each layer’s update and can hold in every layer, but condition (i) holds only for the first: the input of layer 2 is the output of layer 1, which is not constant within a base interval, so a stacked recurrence on the observation grid sees a different layer-2 input after splitting. The resulting change is a reconstruction error of an *internal* signal, not a change of the observed input. Proposition 4.2(b) is an upper bound of order δ for the stated scalar cascade; it does not show that the error is nonzero or of first order for every input (for $u \equiv 0$ it is zero), and it is not a theorem about deep selective networks. Two quantities should be kept apart. The *invariance defect*, the change of the base-time outputs under splitting, vanishes if the inter-layer signal is represented exactly or if the stack runs on a fixed grid that does not depend on the observations (for example the training grid after resampling). The *approximation error* of the layerwise computation relative to the continuous coupled system does not vanish on a fixed grid; for the cascade it is bounded as in Proposition 4.2(b) with the mesh of that grid.

4.3. Readouts

The time average of the output over the held path is additive over sub-intervals, so it is split-invariant whenever the state trajectory is (for example under the conditions of Proposition 4.1); endpoint sums converge to it only where the grid is refined; sample-count readouts weight samples, not time, so the sample mean converges to a density-weighted average. If the first half of $[0, T]$ is split by m and the second half is not, the token mean of outputs that merely repeat the base-grid outputs differs from their base-grid mean by $\frac{m-1}{2(m+1)}(A - B)$, where A and B are the mean base-grid outputs over the two halves ($\frac{7}{18}(A - B)$ for $m=8$); this weighted-average identity involves no property of the

model.

4.4. Interval conventions in the TIDES paper and code

The TIDES paper writes $x_{k+1} = \bar{\Lambda}(\Delta_k)x_k + \bar{B}(\Delta_k)u_k$ with $\Delta_k = t_{k+1} - t_k$ and $y_k = Cx_k + Du_k$, i.e. u_k is held forward on $[t_k, t_{k+1})$ and the state in y_k does not yet contain u_k (Soydan et al., 2026). The pinned code scans $z_k = \bar{\Lambda}(t_k - t_{k-1})z_{k-1} + \bar{B}(t_k - t_{k-1})u_k$ with $y_k = \text{Re}(Cz_k) + Du_k$, and its forecasting collate builds the step sequence $[t_0, t_1 - t_0, \dots]$: u_k is held backward on $(t_{k-1}, t_k]$ and the state in y_k contains it. Both are exact ZOH solutions, but of different input reconstructions. On a uniform grid every gap is equal, so $z_k = x_{k+1}$: the difference is a one-index shift of the state, which changes the alignment between the state part of y_k and the feedthrough Du_k . On an irregular grid each value is paired with a different gap, so the difference is not a relabeling. With the pinned code, untrained weights and a fixed impulse input, the code equals the backward-hold recurrence to 6.5×10^{-19} ; the one-index shift reproduces the paper form to 8.7×10^{-19} on a uniform grid but leaves 3.7×10^{-3} on an irregular one. The released code, not the printed equations, defines the model behind TIDES’ reported results, and it coincides with the right-endpoint convention used throughout this paper; our splitting protocol repeats the right-endpoint value accordingly. We make no claim that this affects TIDES’ results.

5. Trained Official TIDES: Fixed-Protocol Seed Repetition

Setup. *Model:* official TIDES at a pinned commit (MIT), 23,397 parameters, two blocks, input-dependent $\text{Re } \Lambda$, B , C , exact ZOH, causal, no convolution, a linear per-step head; CPU, float32, 2 threads. *Task:* regression of the teacher (1) with fixed parameters, driven by sine-sum inputs, over $T=8$ on a uniform training grid of $K=32$ steps; labels are the teacher output on the continuous path at the 32 grid times (label s.d. 2.35); trajectory identifiers are split into 512/128 train/calibration before any grid is built, with normalization from the training split only. *Training:* Adam, 2000 steps, batch 32; calibration MSE every 100 steps; the lowest-calibration-MSE state is saved, hashed and frozen before any test trajectory is generated. A development run with seed 0 (Appendix C) was used to choose the questions below. The repetition was then fixed in advance: training seeds 1–4, each seeding the parameter initialization and the minibatch order but not the data; the train and calibration data are hash-identical to the development run, and retraining seed 0 with the repetition code reproduces every tensor of the development checkpoint (maximum absolute difference 0). All seeds share one new test set of 256 trajectories from the same generator, with identifiers disjoint from all earlier ones; no seed was replaced or added. This

is a repetition on the same task, not a new task.

Conditions, arms and estimands. The held-path refinements are C0 (training grid), S2/S4/S8 (every interval split with the held value repeated) and H8 (only the first half split by 8, a density change on the same held path); J1 (K random observation times) changes the observed path and is reported separately. *Native* evaluation feeds the condition’s grid with physical step sizes; *resampled* evaluation carries the last observation forward onto the training grid using only observed times and values. On S and H conditions resampling reproduces the C0 input exactly, so its zero discrepancy is an identity control, not evidence of superiority. For a sequence-level readout let $\bar{y}_i = T^{-1} \int_0^T y_i dt$ be the pooled label (Simpson’s rule on the teacher’s RK4 nodes) and $\hat{y}_{i,k}(c)$ the native outputs on condition c with steps $dt_k(c)$ and L_c tokens. We compare the token mean $p_i^{\text{mean}}(c) = L_c^{-1} \sum_k \hat{y}_{i,k}(c)$, the pooling of the TIDES classifier head (here applied to the outputs of a linear per-step head, which equals pooling the features first), with the time-weighted sum $p_i^{\text{time}}(c) = T^{-1} \sum_k \hat{y}_{i,k}(c) dt_k(c)$ of the *same* outputs. The latter is a pooling-only ablation, not part of the official model and not an exact integral; the resampled arm pools its K outputs by $p_i^{\text{res}}(c) = K^{-1} \sum_k \hat{y}_{i,k}^{\text{res}}(c)$. On the uniform C0 grid all three equal $p_i(\text{C0})$; the pooled error of an arm is $n^{-1} \sum_i |p_i(c) - \bar{y}_i|$. The primary estimand, chosen after the development run, is the pooled-readout effect $D_s = \frac{1}{n} \sum_i (|p_{s,i}^{\text{mean}}(\text{H8}) - p_{s,i}(\text{C0})| - |p_{s,i}^{\text{time}}(\text{H8}) - p_{s,i}(\text{C0})|)$ in label units. Secondary estimands are the final-output discrepancy under S8, $\|\hat{y}(\text{S8}) - \hat{y}(\text{C0})\|_2 / \|\hat{y}(\text{C0})\|_2$ over the 32 grid times (also with weights and inputs cast to float64); the loss contrast $r_s = (\bar{L}_{\text{native}} - \bar{L}_{\text{resampled}}) / \bar{L}_{\text{resampled}}$ of the per-trajectory MSE; and wall time. Uncertainty is a paired bootstrap over the 256 test trajectories with the seed fixed, with ratios recomputed in every draw. Because the seeds share the trajectories (a crossed design), we report the seeds separately and give only their range, not a pooled interval. A loss difference is called negligible only if the interval of r_s lies inside $\pm 5\%$, a convention (about 2.5% in RMSE) rather than an application tolerance; non-significance is never read as equivalence.

Model coupling. The final-output discrepancy under S8 is 4.3×10^{-3} –0.022 in the four seeds, with every interval excluding zero (Table 1); it varies 5.1-fold between seeds, and the seed whose checkpoint was selected earliest (step 800) has the smallest value, a relation we did not test. As a precision diagnostic for these checkpoints, casting weights and inputs to float64 changes the mean discrepancy by at most 5.2×10^{-6} (relative 3.8×10^{-4}) and any per-trajectory value by at most 1.8×10^{-5} , while the smallest per-trajectory discrepancy is 1.6×10^{-3} (all from the stored double-precision values of the de-normalized outputs); the discrepancy is therefore not a float32 rounding effect of

Table 1. Model-coupling effect (P1-REAL-02, training seeds 1–4; same task, data and checkpoint rule; shared test set of 256 trajectories, ids 10000–10255). Step and calib. MSE: selected checkpoint and its calibration MSE (label units squared). S8 disc.: mean over trajectories of the relative ℓ_2 change of the final per-step outputs at the 32 training-grid times when every held interval is split into 8 sub-steps, against C0 (float32). Brackets: 95% paired bootstrap CI over test trajectories with the seed fixed.

seed	step	calib. MSE	S8 disc. [95% CI]
1	1300	0.0127	0.0113 [0.0106, 0.0121]
2	1600	0.0135	0.0219 [0.0203, 0.0236]
3	800	0.0168	0.0043 [0.0041, 0.0045]
4	1800	0.0125	0.0147 [0.0140, 0.0153]

these checkpoints. The pre-registered threshold 10^{-4} came from a scalar float32 emulation (Appendix D) and is not a numerical floor of a trained 256-token network.

Pooled weighting. The primary effect is positive with an interval excluding zero in all four seeds (D_s between 0.578 and 0.602, smallest lower bound 0.512; Table 2), so the pre-specified replication criterion within this task is met. Its cause, however, is mostly the readout’s weights rather than the model. Per trajectory the token-mean change splits exactly into the density-weighting term $W = \frac{7}{18}(A - B)$, computed from the unchanged C0 outputs (Section 4.3), and a model term M from the native H8 outputs. On average the change is +0.46 to +0.50, W is +0.48 to +0.49 ($A - B$ is 1.22–1.26, i.e. the C0 outputs of this task are larger in the first half) and M is −0.016 to +0.011; W has the sign of the change in at least 98% of trajectories, and token-mean pooling of outputs that merely repeat C0 would err by 0.56–0.59 against the pooled label, about as much as the native token mean (0.57–0.59). The H8 token-mean effect is therefore a property of sample-count pooling applied to a non-constant output, which any per-token model would show; it is not evidence of a failure specific to selective SSMs. The model-dependent part is small: $|M|$ averages 0.023–0.028, and 0.0005–0.0049 on the tokens that end a base interval, where it measures the change of the model output itself; on the other tokens it also contains the evolution of the output within an interval, which is not an artifact. Time weighting of the same native outputs reduces the mean absolute change against C0 from 0.60–0.62 to 0.013–0.023 but does not remove it, and its pooled error (0.068–0.094) stays close to that of the resampled arm (0.066–0.096). Per-token outputs were not stored, so W , M and Q come from one additional frozen forward pass per checkpoint, which reproduces the stored pooled values exactly (Appendix A). The signed terms are not an additive decomposition of D_s , which is a difference of absolute values.

Task risk: the accuracy consequence does not replicate. The development run suggested that native evaluation on

refined grids is slightly more accurate than resampled evaluation. Across seeds 1–4, r_s ranges from −6.2% to +3.1% on S8 and from −4.9% to +6.7% on H8 (Table 3): seed 2 is worse on H8 with an interval that excludes zero, seed 3 shows no practical difference on every held-path condition, and the remaining verdicts are mixed. We therefore report no stable accuracy difference between native and resampled evaluation on held-path refinements. On J1, native evaluation has lower MSE in every seed on the new test set (r_s from −22.6% to −21.2%), but the development checkpoint, which gives −20.9% here, gave −6.4% (inconclusive) on its own test set, so the J1 contrast depends strongly on the test draw; J1 is not a refinement of the held path.

Cost. With the pre-specified timing protocol (inference mode, two warm-up and five timed calls, one batch of 256 sequences, 2 threads), the ratio native.S8_end_to_end/resampled.S8_end_to_end is 9.7–11.5 across seeds and native.S8_forward/resampled.S8_forward is 11.5–14.1; on C0 the end-to-end ratio is 0.98–1.04 (Table 6). End-to-end time includes resampling and tensor construction; for the resampled arm on S8 the difference between the end-to-end and forward-only medians is 0.007–0.019 s. The two boundaries are timed in separate jobs, and in 4 of 60 seed-condition-arm cells the end-to-end median is below the forward-only median (by up to 9.5%), so timing differences of a few percent are within run-to-run variation; the order-of-magnitude S8 ratio is not. The development run’s timings (1.07 s native on S8) were forward-only, without warm-up and outside inference mode, and excluded resampling; they are superseded for cost statements.

Where the discrepancy enters (exploratory, development checkpoint). After seeing the development results we measured the change at intermediate features of the development checkpoint (Table 10 in Appendix C); this was not pre-registered. The encoder output is unchanged, the first block changes by at most 7.2×10^{-6} in float32 and 1.8×10^{-14} with the weights cast to float64, and the second block changes by 4.1×10^{-3} to 7.2×10^{-3} in both precisions. This is qualitatively consistent with Propositions 4.1 and 4.2: the first block receives the held input and uses an exact held-coefficient update, whereas the second block receives the first block’s output, which splitting samples at more points within each interval; neither proposition is a statement about this trained network. The repetition pre-specified a per-seed layer diagnostic only if some seed’s S8 discrepancy fell below 10^{-3} ; this did not happen, so the localization rests on one checkpoint.

What this does not show. Four training seeds of one two-block model on one synthetic task give a replication within that task, not evidence about other data, sensors, models or tasks, about TIDES’ benchmarks, about selective

Table 2. Readout decomposition under the same-path density change H8 (first half of $[0, T]$ split by 8; 144 tokens), training seeds 1–4, label units, means over the 256 test trajectories. D_s (primary, chosen after the development run): mean of $|p^{\text{mean}}(\text{H8}) - p(\text{C0})| - |p^{\text{time}}(\text{H8}) - p(\text{C0})|$. Per trajectory, exactly, $p^{\text{mean}}(\text{H8}) - p(\text{C0}) = W + M$ and $p^{\text{time}}(\text{H8}) - p(\text{C0}) = Q$ with $W = \sum_j (a_j - w_j) b_j = \frac{7}{18} (A - B)$, $M = \sum_j a_j (z_j - b_j)$, $Q = \sum_j w_j (z_j - b_j)$; b_j is the C0 output of token j ’s base interval, z_j the native H8 output, $a_j = 1/144$, $w_j = dt_j/T$, and A, B are the mean C0 outputs over the first and second half of $[0, T]$. W uses only C0 outputs: it is the same for any model whose H8 outputs repeat the C0 output within each base interval. The signed terms are not an additive decomposition of the absolute-value estimand D_s . Brackets: 95% paired bootstrap CI (seed fixed).

seed	D_s [95% CI]	$p^{\text{mean}} - p(\text{C0})$	W	M [95% CI]	Q [95% CI]	$ Q $
1	0.578 [0.512, 0.649]	+0.462	+0.476	-0.014 [-0.018, -0.010]	-0.012 [-0.015, -0.009]	0.019
2	0.584 [0.517, 0.655]	+0.473	+0.488	-0.016 [-0.019, -0.012]	-0.015 [-0.017, -0.012]	0.023
3	0.602 [0.536, 0.673]	+0.488	+0.491	-0.003 [-0.007, +0.001]	-0.001 [-0.003, +0.001]	0.013
4	0.602 [0.536, 0.672]	+0.497	+0.486	+0.011 [+0.008, +0.015]	+0.014 [+0.012, +0.016]	0.018

Table 3. Task-risk effect: loss contrast $r_s = (\bar{L}_{\text{native}} - \bar{L}_{\text{resampled}}) / \bar{L}_{\text{resampled}}$ in percent (per-trajectory MSE at the 32 training-grid times; numerator and denominator recomputed in every bootstrap draw), training seeds 1–4. Verdict: = CI inside $\pm 5\%$; < / > native lower / higher (CI excludes 0); << / >> also beyond the margin; ? inconclusive. The $\pm 5\%$ margin is a convention. S2, S8 and H8 keep the held path; J1 (random observation times) changes the observed path. Timing: Table 6.

seed	$r_s(\text{S2})$ [95% CI]		$r_s(\text{S8})$ [95% CI]		$r_s(\text{H8})$ [95% CI]		$r_s(\text{J1})$ [95% CI]	
1	-4.3 [-5.8, -2.7]	<	-6.2 [-8.8, -3.4]	<	-4.9 [-7.2, -2.3]	<	-21.9 [-32.0, -8.8]	<<
2	-2.1 [-5.3, +1.3]	?	+3.1 [-2.5, +9.4]	?	+6.7 [+1.8, +11.9]	>	-21.2 [-31.7, -6.5]	<<
3	-1.2 [-2.4, -0.2]	=	-1.9 [-3.9, -0.1]	=	-0.6 [-1.7, +0.6]	=	-21.8 [-31.9, -8.9]	<<
4	-5.4 [-8.0, -2.8]	<	-4.5 [-9.6, +0.7]	?	-2.1 [-5.7, +1.9]	?	-22.6 [-32.9, -9.0]	<<

SSMs in general, or a precise statement about the distribution of training runs. The teacher is from the model’s own continuous-time family, the primary estimand was chosen after the development run, and the calibration curves were unstable in every seed (up to 0.84 in the development run), so the selected checkpoints depend on the evaluation schedule.

6. Controlled Toy Results

These fixed-parameter results are numerical checks and controlled measurements, not proofs, and say nothing directly about trained models. The pre-registered FIRST.RUN used 32 independent units (one parameter and input draw each), $T=8$, $K=16$, $m \in \{1, 2, 4, 8\}$, 2 CPU threads and 20.6 s; details and deviations are in Appendix A.

Splitting. With $\Delta=dtg$ and exact ZOH the output is split-invariant to 6.3×10^{-15} and agrees with an independent RK4 reference to 1.6×10^{-13} (an engineering check of Proposition 4.1). Euler- B with the correct step converges at first order (median slope -1.03) but has a median base-step artifact of 0.21 in this toy; the magnitude depends on the toy’s $\Delta|\lambda|$ scale. Bilinear converges at second order (-2.02); with $\Delta=\tau g$ the artifact grows from 0.48 to 1.3 (medians, $m=2 \rightarrow 8$), by construction (Table 12, Figure 1a). In a stiff mode ($\lambda = -500$) the bilinear transition is -0.977 and Euler- B has an artifact of 170 (Appendix E).

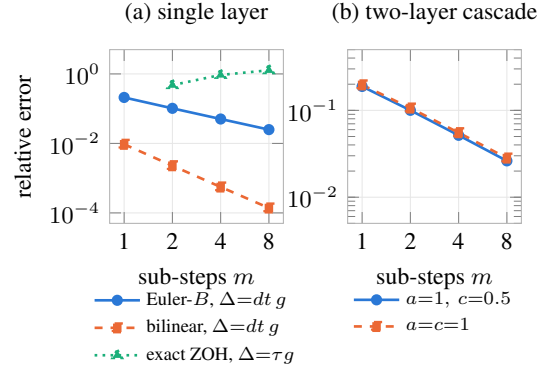


Figure 1. Error against the exact held-path solution as the same held interval is split into m sub-steps: (a) single layer (FR-E1-SPLIT; medians over 32 toy units); (b) two-layer cascade (P1-COMP-01; one deterministic case per line). Exact ZOH with $\Delta=dtg$ (a) and the exact coupled solution (b) stay at 10^{-15} and are not drawn. The $\Delta=\tau g$ line starts at $m=2$ because it is exact at $m=1$ by construction. Values are listed in Tables 12 and 13.

New observations. With exact ZOH the artifact against RK4 on the new held path stays below 1.7×10^{-13} , so the whole change is the change of the continuous-time solution on the new path, i.e. a response to the changed input reconstruction. The exact decomposition (2) of the other variants (exploratory, Appendix E) has cross terms between -1.37 and $+0.42$ of the per-unit total, so no per-unit share is defined.

Table 4. One fix at a time (FR-E3-READOUT, split of the first half, $m=8$; median scale-normalized readout change over 32 units). The effects are not additive, so no per-fix share is reported.

fixes applied	change
none ($\Delta=\tau g$, Euler- B , sample mean)	0.46
time readout only (trapezoidal)	0.25
exact ZOH B only	0.48
$\Delta=dt g$ only	0.30
$\Delta=dt g$ + trapezoidal	0.049
$\Delta=dt g$ + exact ZOH (sample mean)	0.30
$\Delta=dt g$ + exact ZOH + trapezoidal	0.011
$\Delta=dt g$ + exact ZOH + exact integral	1.5×10^{-16}

Readouts and interaction of fixes. On a split-invariant backbone, a density change moves the sample mean by 0.30 and the exact integral by 1.5×10^{-16} (scale-normalized, $m=8$). The pre-registered clause H7c was mis-specified before the run and failed (slope +0.42); its restatement passed only weakly (median slope -0.076 , 21 of 32 units negative). One fix at a time (Table 4): exact ZOH alone gives 0.48, the physical step alone 0.30, a time readout alone 0.25, against 0.46 without fixes; in this tested ablation the three corrections combined reduced the artifact to 1.5×10^{-16} . This is not a necessity result: it concerns a time-average readout of one toy, a last-state readout needs no time integral, and special inputs or coefficients (for example a zero input) remove the artifact without all three corrections. Under the pre-registered stop condition this means the toy gives no support for a new architecture; the stop condition is an operational rule, not a statistical, equivalence or novelty test.

Two-layer cascade. For (3) (P1-COMP-01; held input of FIRST.RUN unit 0; 0.11 s), the exact coupled solution is split-invariant to 1.7×10^{-15} , while the layerwise computation has error 0.19 at the base step and 0.026 at $m=8$ (slope -0.95 ; -0.93 for $a = c$; Figure 1b, Table 13), consistent with, not a proof of, first order for this input.

7. Limitations

Trained evidence: four training seeds and a development seed of one small two-block model (one selective SSM) on one synthetic task whose teacher is from the model’s own family; no statement about other tasks, real data, the population of training runs, TIDES’ benchmarks, Mamba, S4 or S5 is supported, and the accuracy contrast was not stable. A first evaluation on real sensor signals has not been run. *Margin:* the $\pm 5\%$ equivalence margin is a convention, not an application-derived tolerance; all intervals are reported. *Entry points not measured:* token-indexed convolution, normalization in training mode, and deeper or wider stacks. *Analysis:* Propositions 4.1 and 4.2 are standard; Proposition 4.2 is an upper bound for a linear time-invariant scalar cascade (Remark 4.3) and gives no lower bound and no state-

ment about deep selective networks; no global hold-error or Euler- B bound is claimed. *Post hoc choices:* the primary estimand of the repetition was chosen after the development run; the layer localization, the H8 signed decomposition, the toy decomposition and the stiffness columns were analysed after the results were seen. *Numerics:* the 10^{-4} float32 threshold comes from a scalar emulation; for the trained checkpoints we rely on their own float32/float64 comparison. *Literature:* the key equations and experiment sections of TIDES, Mamba-3 and S5 were read in the original text; other entries were checked by keyword search.

8. Conclusion

Grid dependence in selective SSMs can enter through a step that counts samples, an inexact input-matrix discretization, sample-count readouts and, in stacks, the resampling of internal signals; the last is not caused by any change of the observed input. In the tested single-layer toy ablation, the combined corrections (physical step, exact ZOH, exact time average) reduced the readout artifact to rounding level while no single correction did; the inter-layer resampling error remained, with an $O(\text{mesh})$ upper bound for a scalar linear cascade. On the official TIDES code, whose step is physical, splitting the held path changes the final outputs in all four training seeds of one task; the larger readout effect of a same-path density change is almost entirely the weighted-average identity of token-mean pooling, which time weighting reduces to a small residual; the loss consequence of native versus resampled evaluation does not replicate, and resampling is about ten times cheaper at eight-fold refinement. Whether these effects matter on real sensor data and for other selective SSMs is open.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the understanding of how sequence models respond to changes in sampling. All experiments use synthetic data and small CPU computations; the trained-model results come from four training seeds and a development seed on one synthetic task. The main risk we see is over-reading such results: a model that behaves consistently in a controlled setting may still behave differently on irregularly sampled real data, for example clinical or sensor streams, so any use in such settings needs evaluation on the actual data and sampling process. We are not aware of other societal consequences that must be specifically highlighted here.

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A. Pre-registration, deviations and compute

FIRST_RUN. Hypotheses H1–H8, metrics, 32 units, 2 CPU threads, a 120 s budget and stop conditions were committed before any experiment code ran. Amendments recorded before any experiment ran: A1 (adaptive RK4 reference step), A2 (a restated H7c, keeping the mis-specified original), A3 (secondary configurations fixed to seeds 0–7, descriptive). Deviations: a timing-only smoke test on seed 999 (metrics not inspected); a repeated execution after a manifest bookkeeping bug, with byte-identical data files; a partial execution killed by a shell pipe, whose outputs were deleted. P1-COMP-01 was pre-registered separately before its code ran. Table 5 lists every pre-registered verdict, including the failure.

P1-REAL-01. The design, the pinned implementation and the predictions were committed before any installation. Amendments R-A1–R-A9 were committed after an untrained adapter check and a timing smoke on four training trajectories, and before training: calibration-based checkpoint selection, same-held-path refinements only (new-observation conditions removed, one random-time condition kept as secondary), a resampler restricted to observed values, the pooled readout and pooling-only arm, the explicit equivalence unit, the single-seed pilot label, time caps, and seeding of all random generators (the TIDES initialization also draws from NumPy’s global generator). The pre-registered predictions were: native splitting discrepancy at S8 above 10^{-4} (supported), resampled splitting discrepancy exactly zero (supported), native runtime growing with tokens (observed), and a larger sample-mean than time-weighted pooled change on H8 (supported).

P1-REAL-02. The configuration (seeds, identical protocol, new test set, primary and secondary estimands, bootstrap and margin rules, timing boundaries, failure handling, conditional extra diagnostics) was committed before any repetition run. One incident occurred: the evaluation of the development row crashed before writing any output because its output directory was not created; the attempt is recorded as INVALID, the directory creation was added, and only that row was rerun (the four training seeds were unaffected). The pre-specified conditional layer diagnostic was not triggered.

H8 signed decomposition (derived, after the results). P1-REAL-02 stored per-trajectory pooled values but not per-token outputs. A configuration fixed before execution specified one frozen native forward pass per existing checkpoint on C0 and H8 (no training, no timing, same test set), a check that it reproduces the stored pooled values (maximum absolute difference 0), and the signed identities of Table 2 (largest residual 4.6×10^{-15}).

Compute. All on CPU (2 threads for experiments), no GPU, no paid API, no external data; wall and process CPU seconds are distinguished where recorded. FIRST_RUN: two executions of about 21 s, one partial execution and a smoke test of about 3 s together. P1-COMP-01: 0.11 s. P1-REAL-01 (development): installation of a CPU tensor library (35 s) and of a $\text{\LaTeX}$ toolchain (66 s); adapter check and timing smoke (≈ 4 s; 0.031 s per batch-32 step); convention check (≈ 2 s); training including label generation (97 s); evaluation (24 s); layer diagnostic (≈ 11 s). P1-REAL-02: data generation 40 s; seed-0 equivalence retrain ≈ 57 s; four training runs 188 s wall in total (364 s process CPU in the training loops); test-set generation 16 s; five evaluations 143 s wall in total, plus the invalid development attempt (≈ 30 s); H8 decomposition forward passes 7.7 s.

Table 5. Pre-registered check verdicts, including the failed H7c_original (mis-specified before the run; amendment A2) and the weakly supported H7c_restated. These are bookkeeping, not scientific results.

run	check	verdict
FIRST_RUN	H1	PASS
FIRST_RUN	H2	PASS
FIRST_RUN	H3	PASS
FIRST_RUN	H4	PASS
FIRST_RUN	H5a	PASS
FIRST_RUN	H5b	PASS
FIRST_RUN	H6	PASS
FIRST_RUN	H7a	PASS
FIRST_RUN	H7b	PASS
FIRST_RUN	H7c_original	FAIL
FIRST_RUN	H7c_restated	PASS (weak; see text)
FIRST_RUN	H8	PASS
P1-COMP-01	COMP-H1 (distinct)	PASS
P1-COMP-01	COMP-H2 (distinct)	PASS
P1-COMP-01	COMP-H3 (distinct)	PASS
P1-COMP-01	COMP-H4 (distinct)	PASS
P1-COMP-01	COMP-H5 (distinct)	PASS
P1-COMP-01	COMP-H1 (equal)	PASS
P1-COMP-01	COMP-H2 (equal)	PASS
P1-COMP-01	COMP-H3 (equal)	PASS
P1-COMP-01	COMP-H4 (equal)	PASS
P1-COMP-01	COMP-H5 (equal)	PASS

B. Implementation and protocol details

Pinned code facts (read, then executed through the adapter): the step is $\text{step_scale} * \exp(\log_step)$ with a per-mode timescale; the ZOH input matrix is $(1/\text{Lambda}) * (\text{Lambda_bar}-1) * B$ without `expm1`; the scan is $x_k = \bar{\Lambda}_k x_{k-1} + \bar{B}_k u_k$ with the gap before step k ; each block applies `BatchNorm1d(affine=False)` over batch and time (running statistics at evaluation); an optional causal convolution defaults to off; the classifier head pools with `h.mean(dim=1)`. *Adapter check (untrained, four training trajectories):* forward outputs finite; gradients finite for all 54 parameters that receive one (the six static Re Λ , B , C tensors are unused in input-dependent mode; parameters upstream of the zero-initialized head projections receive zero gradient at step 0 only); an Adam step changed the parameters and lowered the loss on the same batch; evaluation was deterministic, and a per-step step tensor of ones matched the scalar step exactly. *Labels:* teacher output on the continuous path from an independent RK4 integration (128 sub-steps per training interval); the pooled label is its time average by Simpson’s rule. *Readout of J1:* the last native output at or before each training-grid time. *Eval-mode checks (P1-REAL-02, every seed):* every module in evaluation mode during calibration and test; dropout probability 0; batch normalization with running statistics; equal sequence lengths within every batch (no padding or masking is used); all step sizes positive (the smallest native step is 1/8 of the training step on S8 and H8). *Timing (P1-REAL-02):* Table 6.

Table 6. P1-REAL-02 timing (one batch of 256 sequences; median of 5 timed calls after 2 warm-ups; `inference_mode`; 2 threads). End-to-end includes tensor construction, LOCF resampling for the resampled arm, forward, de-normalization and readout mapping; forward-only is the model call on prebuilt tensors. CPU seconds are process CPU time summed over threads. Timed repeats are not independent samples. [‡]Development checkpoint (seed 0).

seed	cond.	tokens	nat. e2e	nat. fwd	res. e2e	res. fwd	nat. e2e CPU
0 [‡]	C0	32	0.049	0.044	0.047	0.042	0.10
0 [‡]	S8	256	0.578	0.574	0.068	0.063	1.12
0 [‡]	H8	144	0.297	0.281	0.069	0.056	0.58
1	C0	32	0.042	0.039	0.042	0.041	0.09
1	S8	256	0.565	0.545	0.049	0.041	1.10
1	H8	144	0.308	0.275	0.047	0.042	0.60
2	C0	32	0.041	0.038	0.042	0.038	0.08
2	S8	256	0.560	0.550	0.058	0.039	1.09
2	H8	144	0.300	0.281	0.053	0.043	0.59
3	C0	32	0.051	0.044	0.049	0.043	0.10
3	S8	256	0.544	0.566	0.054	0.047	1.06
3	H8	144	0.286	0.275	0.061	0.047	0.56
4	C0	32	0.045	0.041	0.045	0.042	0.09
4	S8	256	0.574	0.531	0.054	0.046	1.10
4	H8	144	0.281	0.276	0.055	0.047	0.56

C. Development pilot (seed 0) on its own test set

Table 7 gives the development checkpoint on the new shared test set. The remaining results come from the development run (P1-REAL-01), whose questions and estimands were not fixed in advance beyond its own predictions, on its own test set (identifiers 640–895). Its equivalence analysis used a margin of 5% of the point-estimated resampled MSE (kept here as exploratory); the repetition in Section 5 instead bootstraps the ratio r_s . Pooled quantities in Table 9 are mean absolute errors (or changes) of pooled predictions against the pooled label, in label units; they are not hidden-state discrepancies or MSEs. Its runtimes are forward-only model calls on prebuilt tensors (no warm-up, outside inference mode) and exclude resampling.

Table 7. Development checkpoint (seed 0, P1-REAL-01) evaluated with the P1-REAL-02 protocol on the new shared test set; not part of the replication criterion. Definitions as in Tables 1 to 3.

quantity	development checkpoint, new test set
selected step / calibration MSE	1500 / 0.0119
S8 disc. [95% CI]	0.0114 [0.0108, 0.0120]
D_s [95% CI]	0.575 [0.511, 0.643]
$p^{\text{mean}} - p(\text{C0}), W, M, Q$ (means)	+0.452, +0.460, −0.009, −0.009
$r_s(\text{S2})$ [95% CI], verdict	−2.9 [−4.6, −1.5] =
$r_s(\text{S8})$ [95% CI], verdict	−3.9 [−6.7, −1.8] <
$r_s(\text{H8})$ [95% CI], verdict	−6.5 [−9.4, −3.8] <
$r_s(\text{J1})$ [95% CI], verdict	−20.9 [−30.8, −8.0] <<

Table 8. DEVELOPMENT run (P1-REAL-01): trained official TIDES, seed 0, on its own test set (256 trajectories, ids 640–895; one checkpoint selected on the calibration split; paired). “disc.”: mean relative ℓ_2 change of the per-step outputs at the 32 training-grid times against the same arm on C0. MSE: per-trajectory mean squared error to the teacher at those times (original units; label s.d. 2.35). Δ MSE: mean paired difference native – resampled in units of 10^{-4} (label units squared) with a 95% bootstrap CI; the margin is $\pm 5\%$ of the point-estimated resampled mean MSE of the same condition (exploratory rule of the development run). Verdict (native vs. resampled MSE): lower/higher if the CI excludes 0 on that side. Runtimes of this run are not tabulated; they were forward-only and are superseded by Table 6. C0 is the training grid; S m and H8 (first half split by 8) keep the held path; † J1 (random observation times) is a secondary non-refinement condition whose held path differs by construction.

cond.	disc. native	disc. resampled	MSE native	MSE resampled	Δ MSE [95% CI] (10^{-4})	margin rule
C0	0	0	0.0167	0.0167	0 [0, 0]	identical inputs (check)
S2	6.1×10^{-3}	0	0.0162	0.0167	-5.1 [-7.3, -2.7]	within $\pm 5\%$
S4	9.2×10^{-3}	0	0.0161	0.0167	-6.5 [-9.9, -2.9]	lower
S8	0.011	0	0.0160	0.0167	-6.9 [-10.8, -2.7]	lower
H8	7.8×10^{-3}	0	0.0156	0.0167	-10.8 [-14.0, -7.6]	lower
J1 †	0.35	0.39	1.14	1.22	-785 [-2120, +623]	inconclusive

Table 9. Pooled readout of the development checkpoint on its own test set (P1-REAL-01), label units, means over 256 test trajectories. Left: $n^{-1} \sum_i |p_i(c) - \bar{y}_i|$; right: $n^{-1} \sum_i |p_i(c) - p_i(C0)|$, with $p = p^{\text{mean}}$ (“native mean”, token mean as in the TIDES classifier head), $p = p^{\text{time}}$ (“native time”, right-endpoint time-weighted sum of the same native outputs; a pooling-only ablation, not an exact integral) or $p = p^{\text{res}}$ (“resampled”, token mean of the 32 resampled outputs); definitions in Section 5. Pooled label $\bar{y}_i$: time average of the teacher output over $[0, T]$ by Simpson’s rule on its RK4 nodes (mean absolute value 1.59).

cond.	error to pooled label			change vs. C0		
	native mean	native time	resampled	native mean	native time	resampled
C0	0.063	0.063	0.063	0	0	0
S8	0.070	0.070	0.063	0.032	0.032	0
H8	0.56	0.068	0.063	0.55	0.016	0
J1	0.37	0.28	0.29	0.36	0.26	0.30

Table 10. Where the splitting change appears in the development checkpoint (EXPLORATORY, added after its results were seen). Mean relative ℓ_2 change against C0 at the training-grid times of the features after each block and of the (normalized) output, with the frozen weights evaluated in float32 and cast to float64. The encoder output is unchanged (0) in all cases.

cond.	block 1		block 2		output	
	fp32	fp64	fp32	fp64	fp32	fp64
S2	5.6×10^{-6}	5.1×10^{-15}	4.1×10^{-3}	4.1×10^{-3}	6.8×10^{-3}	6.8×10^{-3}
S4	6.0×10^{-6}	1.1×10^{-14}	6.1×10^{-3}	6.1×10^{-3}	0.010	0.010
S8	7.2×10^{-6}	1.8×10^{-14}	7.2×10^{-3}	7.2×10^{-3}	0.012	0.012
H8	5.8×10^{-6}	1.4×10^{-14}	5.5×10^{-3}	5.4×10^{-3}	9.1×10^{-3}	9.1×10^{-3}

D. Numerical floor

The naive coefficient $(e^z - 1)/\lambda$ loses relative precision as $|z|$ shrinks (8.0×10^{-4} at $z = -10^{-14}$ in float64), while `expm1` stays at rounding level (Table 11). In emulated float32, split composition of one scalar mode drifts to 1.1×10^{-4} at 16,384 sub-steps, so differences below about 10^{-4} are not interpretable in that emulation; this is not a floor for a trained multi-layer network with 256 tokens, for which Section 5 compares float32 with float64 directly.

Table 11. Numerical floor (FR-E4). Left: relative error of the ZOH input coefficient $(e^z - 1)/\lambda$ against a 50-digit reference. Right: splitting $[0, 1]$ into m held sub-steps ($\lambda = -1$, $g = \ln 2$, exact $h(1) = 0.5$); float32 is an *emulation* (every primitive rounded, correctly rounded `exp`), not a GPU kernel.

z	naive	<code>expm1</code>	m	fp64 <code>expm1</code>	fp32 <code>expm1</code>	fp32 naive
-0.01	5.4×10^{-15}	4.7×10^{-17}	1	0	0	0
-1×10^{-4}	1.3×10^{-14}	3.4×10^{-17}	4	2.2×10^{-16}	0	6.0×10^{-8}
-1×10^{-6}	1.6×10^{-11}	7.9×10^{-17}	16	2.2×10^{-16}	1.2×10^{-7}	2.4×10^{-7}
-1×10^{-8}	1.1×10^{-9}	6.4×10^{-17}	64	4.4×10^{-16}	6.0×10^{-8}	0
-1×10^{-10}	8.3×10^{-8}	3.4×10^{-17}	256	1.8×10^{-15}	4.3×10^{-6}	5.0×10^{-6}
-1×10^{-12}	2.2×10^{-5}	2.4×10^{-17}	1024	4.4×10^{-15}	1.8×10^{-5}	2.0×10^{-5}
-1×10^{-14}	8.0×10^{-4}	4.9×10^{-17}	4096	1.1×10^{-13}	2.1×10^{-5}	2.8×10^{-5}
			16384	5.2×10^{-13}	1.1×10^{-4}	2.3×10^{-4}

E. Additional toy tables

Table 12. Splitting a held interval into m sub-steps (FR-E1-SPLIT; toy, 32 units, uniform base grid, real modes). Entries are the median over units of the output artifact $\|y_m - y^*\|_2 / \|y^*\|_2$ at the 17 common base times, where y^* is the exact held-path solution; slope is the median per-unit log-log slope over $m \in \{1, 2, 4, 8\}$. τ is the base step, so $\Delta = \tau g$ coincides with $\Delta = dt g$ at $m=1$.

variant	$m=1$	$m=2$	$m=4$	$m=8$	max, $m=8$	slope
exact ZOH, $\Delta=dt g$	0	2.6×10^{-16}	3.8×10^{-16}	5.5×10^{-16}	6.3×10^{-15}	–
Euler-B, $\Delta=dt g$	0.21	0.10	0.050	0.025	0.12	–1.03
bilinear, $\Delta=dt g$	9.5×10^{-3}	2.3×10^{-3}	5.6×10^{-4}	1.4×10^{-4}	8.1×10^{-4}	–2.02
exact ZOH, $\Delta=\tau g$	0	0.48	0.93	1.3	2.8	–
Euler-B, $\Delta=\tau g$	0.21	0.68	1.1	1.4	3.1	0.93
bilinear, $\Delta=\tau g$	9.5×10^{-3}	0.48	0.94	1.3	2.8	2.25

Table 13. Two-layer linear cascade $\dot{h}_1 = -ah_1 + bu$, $\dot{h}_2 = -ch_2 + dh_1$ under interval splitting (P1-COMP-01; $b=d=1$; held input of FIRST_RUN unit 0). Columns m : error of the layerwise computation (layer 2 holds the layer-1 endpoint value) against the exact coupled solution, $\|h_2^{lw} - h_2\| / \|h_2\|$ over base times. “change” is the layerwise change $m=8$ vs. $m=1$; the last column is the largest change of the exact coupled solution over m . No external observation is added in this experiment.

case	$m=1$	$m=2$	$m=4$	$m=8$	slope	change	exact inv.
$a=1, c=0.5$	0.19	0.10	0.052	0.026	–0.95	0.15	1.7×10^{-15}
$a=c=1$	0.20	0.11	0.055	0.028	–0.93	0.16	7.1×10^{-16}

Table 14. Stiff scalar mode (FR-E4-N2; $g = \ln 2$, base step 0.5, held sine input). Bilinear is stable ($|\bar{A}| < 1$) for every step, but for $z = \Delta\lambda < -2$ its transition is negative (sign-alternating transients) and tends to -1 , whereas the exact decay e^z tends to 0. The last two columns are the output artifact against exact ZOH. The stable/sign/decay columns are an exploratory re-reading of logged values.

λ	m	z	$\bar{A}_{\text{bil}}$	e^z	stable	sign-alt.	bilinear	Euler-B
–5	1	–1.73	+0.072	0.18	yes	no	0.098	1.1
–5	4	–0.43	+0.644	0.65	yes	no	4.7×10^{-3}	0.23
–5	16	–0.11	+0.897	0.90	yes	no	2.9×10^{-4}	0.055
–5	32	–0.05	+0.947	0.95	yes	no	7.3×10^{-5}	0.027
–50	1	–17.33	–0.793	3.0×10^{-8}	yes	yes	0.54	16
–50	4	–4.33	–0.368	0.013	yes	yes	0.015	3.4
–50	16	–1.08	+0.297	0.34	yes	no	2.1×10^{-8}	0.64
–50	32	–0.54	+0.574	0.58	yes	no	8.5×10^{-9}	0.30
–500	1	–173.29	–0.977	5.5×10^{-76}	yes	yes	1.1	170
–500	4	–43.32	–0.912	1.5×10^{-19}	yes	yes	0.62	42
–500	16	–10.83	–0.688	2.0×10^{-5}	yes	yes	2.0×10^{-3}	9.8
–500	32	–5.42	–0.461	4.4×10^{-3}	yes	yes	1.3×10^{-11}	4.4

dt	exact ZOH	Euler-B	bilinear
0.01	8.0×10^{-15}	3.5×10^{-3}	0
0.5	0	0.18	1.1×10^{-16}
8	0	4.6	5.6×10^{-6}

Lower block (FR-E4-N3): relative steady-state error under a held unit input, $\lambda = -1$.

Table 15. Adding new observations of the same signal (FR-E2-NEWOBS; toy, 32 units). The change $T = y_m - y_1$ at base times splits exactly as $T = P + R$, where P is the change of the continuous-time solution on the new held path (genuine new external information) and R is the change of the discretization artifact. Hence $\|T\|^2 = \|P\|^2 + \|R\|^2 + 2\langle P, R \rangle$; we report these three terms as fractions of $\sum_{\text{units}} \|T\|^2$ (they sum to 1) and the per-unit range of the cross fraction. P uses the exact-ZOH output as a proxy for the continuous-time held-path solution (validated against RK4 to the bound shown). Median total change is relative to $\|y_1\|$. EXPLORATORY re-aggregation; the cross term shows that no per-unit “share” is well defined.

variant	m	median $\ T\ /\ y_1\ $	$\ P\ ^2$	$\ R\ ^2$	$2\langle P, R \rangle$	cross range
exact ZOH, $\Delta=dt\ g$	2	0.11	artifact $\leq 1.7 \times 10^{-13}$ vs. RK4 (proxy reference)			
exact ZOH, $\Delta=dt\ g$	8	0.19	artifact $\leq 1.7 \times 10^{-13}$ vs. RK4 (proxy reference)			
Euler-B, $\Delta=dt\ g$	2	0.15	0.173	0.567	+0.260	$[-0.93, +0.42]$
Euler-B, $\Delta=dt\ g$	8	0.25	0.198	0.533	+0.269	$[-1.37, +0.42]$
bilinear, $\Delta=dt\ g$	2	0.11	0.849	0.025	+0.126	$[-1.04, +0.34]$
bilinear, $\Delta=dt\ g$	8	0.19	0.886	0.013	+0.101	$[-0.69, +0.29]$
exact ZOH, $\Delta=\tau g$	2	0.42	0.071	1.260	-0.331	$[-1.06, +0.34]$
exact ZOH, $\Delta=\tau g$	8	1.1	0.025	1.223	-0.248	$[-1.02, +0.03]$
Euler-B, $\Delta=\tau g$	2	0.42	0.053	1.225	-0.278	$[-0.72, +0.32]$
Euler-B, $\Delta=\tau g$	8	1.1	0.020	1.202	-0.222	$[-0.75, +0.04]$
bilinear, $\Delta=\tau g$	2	0.42	0.072	1.260	-0.331	$[-1.10, +0.35]$
bilinear, $\Delta=\tau g$	8	1.1	0.025	1.222	-0.247	$[-1.04, +0.03]$

Table 16. Readouts under a density change (FR-E3-READOUT; toy, 32 units). Columns 2–3: scale-normalized change $|r_8 - r_1|/s$ when only the first half of the base intervals is split into 8 held sub-steps (physical path unchanged); s is the RMS base-time output at $m=1$ ($\times K_1$ for the sample sum). Column 4: scale-normalized error to the continuous-time truth when new observations are added to the first half only, $m=1 \rightarrow 8$. Medians over units. The exact held-path integral is defined only for exact-ZOH states.

readout	split, exact ZOH $\Delta=dt\ g$	split, Euler-B $\Delta=\tau g$	new obs., exact ZOH
last sample	8.9×10^{-17}	0.19	0.085→0.099
sample sum	1.5	2.9	—
sample mean	0.30	0.46	0.060→0.28
right-endpoint $\sum y_k dt_k$	0.026	0.24	0.060→0.043
trapezoidal endpoint sum	0.011	0.25	0.039→0.023
exact held-path integral	1.5×10^{-16}	n/a	0.039→0.032

F. Reproducibility

The anonymized supplementary package contains the code (toy, RK4 references, cascade check, TIDES adapter, convention check, runners, diagnostics and the generator that produces every table, figure and number in this paper), the configurations, and the raw records from which the generator works, together with the sha256 of every checkpoint. It does not contain model checkpoints or the third-party TIDES code: the checkpoints are regenerated by the training command listed in the package (a retrain of the development seed reproduced every tensor), and TIDES is obtained at the pinned commit under its own license.